

Nerves and brains

Nerves play an essential part in making animal bodies work. Using tiny electrical pulses, they make muscles contract, enabling movement. They also carry information from sense organs so that animals can react to the world around them. Nerves do this by passing on signals. Simple animals, such as jellyfish, have a mesh of nerve cells throughout their bodies. But in most animals, the nervous system has major highways called nerves and nerve cords, which connect with the brain. In the brain, information is processed and stored, letting animals behave in amazingly complex ways.

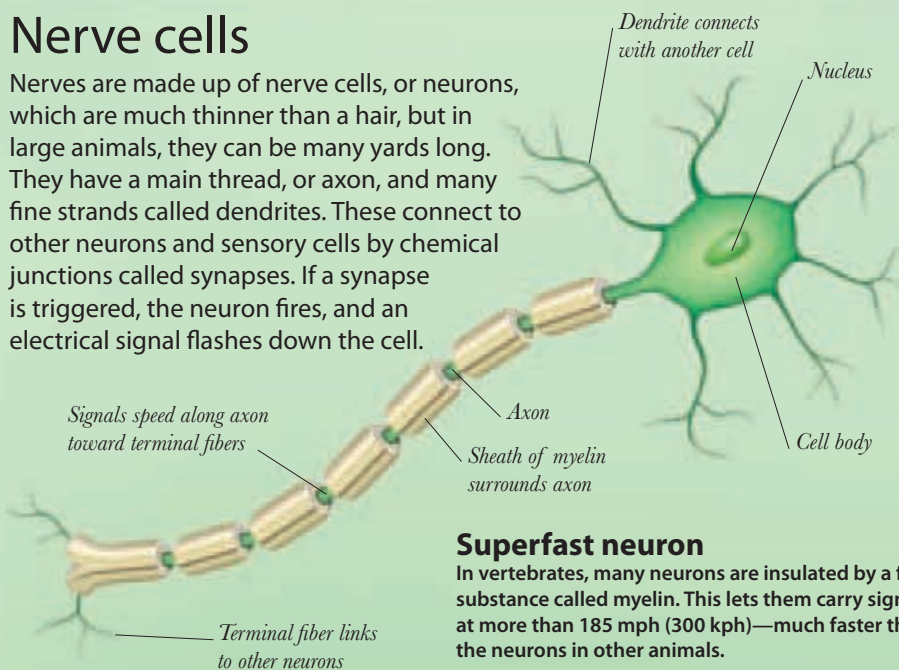


Changing brain

A fox's brain contains more than 100 billion nerve cells, or neurons. Unlike most cells in a fox's body, the neurons are in place when a cub is born. From its very first day, the cub can perform instinctive behavior such as feeding on milk, because this action is programmed into its brain. As the fox grows up, it learns many new kinds of behavior, including how to hunt. As the fox learns new skills and absorbs fresh information, new connections are made between the neurons in its brain.

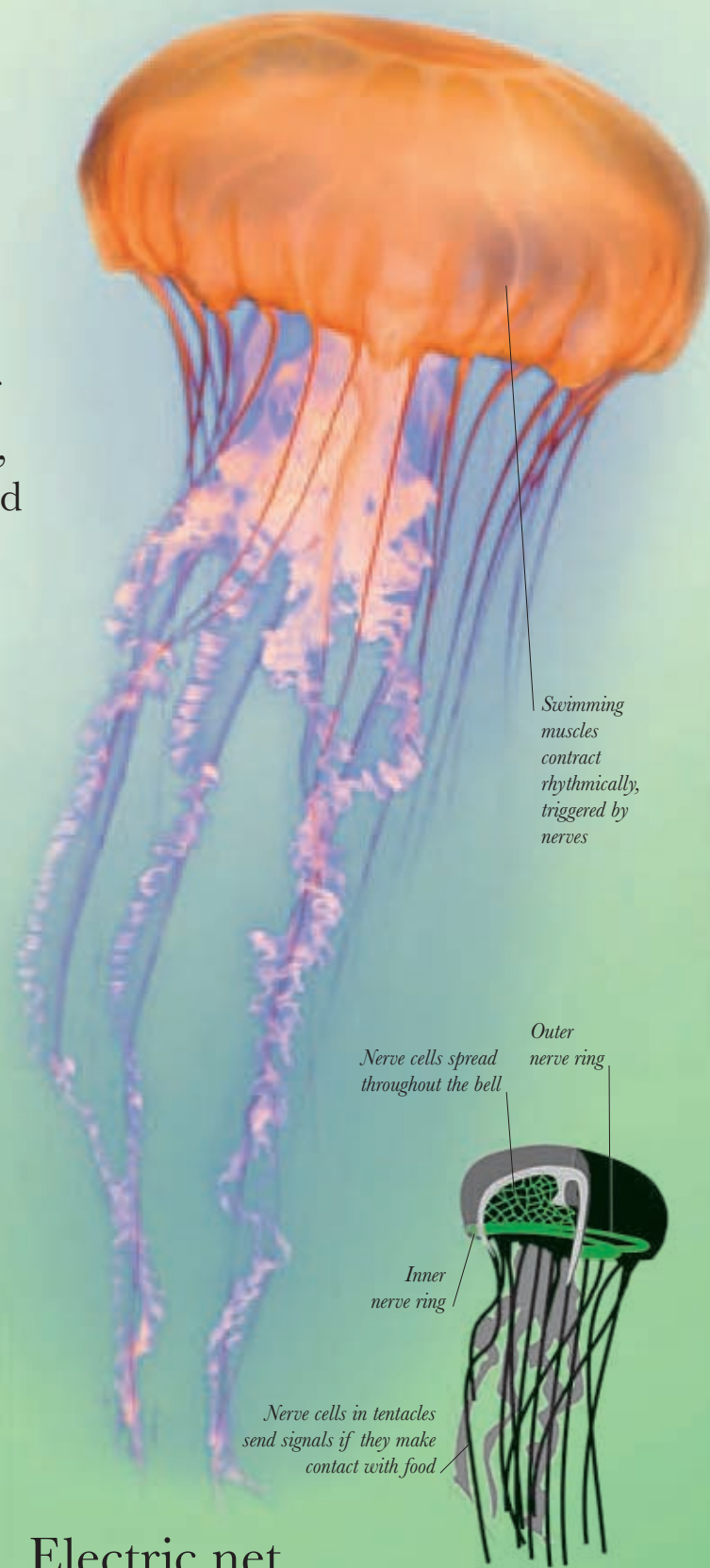
Nerve cells

Nerves are made up of nerve cells, or neurons, which are much thinner than a hair, but in large animals, they can be many yards long. They have a main thread, or axon, and many fine strands called dendrites. These connect to other neurons and sensory cells by chemical junctions called synapses. If a synapse is triggered, the neuron fires, and an electrical signal flashes down the cell.



Superfast neuron

In vertebrates, many neurons are insulated by a fatty substance called myelin. This lets them carry signals at more than 185 mph (300 kph)—much faster than the neurons in other animals.



Electric net

Jellyfish and their relatives have a pattern of nerve cells spread throughout their bodies like a net. Like all nerves, they are connected by synapses, which let signals jump from one nerve cell to the next. Some jellyfish also have a ring of nerves on the inner and outer edges of their bell. The outer ring coordinates the jellyfish's sense of touch, while the inner ring controls swimming, by making the muscles in its bell contract together.